PGR111, tasks for week 08

Task 1 – Concepts & Terminology

Briefly explain the following terminology and database concepts:

- Surrogate key
- Cardinality (for relationships)
- Participation (for relationships)
- Multiplicity (for relationships)
- Super key
- Candidate key

Note: Keep your answers. They might become helpful later in the semester. :-)

Task 2 – ER Modell & SQL

Feel free collaborate with others on the tasks (but you can do the tasks alone as well).
Note: We recommend using paper or **Lucidchart/Gliffy/etc.**, rather than the "data modelling" part of MySQL Workbench: the latter skips (automates) some parts of the tasks, thus you might miss out on some of the intended learning.

Imagine that you have been hired to design a database solution for a university college (like Kristiania). They have the following thoughts on the content. **Create an ER model for part 1-5.**

1. They will store student data in the database, so they easily can send out information, etc. Each student should have a unique student number, as well as the possibility to store a name, an address, and several emails (e.g.: school email, private email, ...).

2. The courses at the school should be stored with an id (6 characters, both numbers and letters) and a title (100 characters).

3. Which field of study each student has chosen must also be registered (each student can only follow one field of study).

4. Each field of study must be linked to the courses it contains. Although it is common to have courses for each field of study, it should also be possible to have common courses for several fields of study.

5. Complete the ER model with the details needed for a working database (add PKs + FKs and replace any M:M relationships with composite tables).

6. Implement the tables in MySQL! (Create a new database/schema: college).

7. Register the following courses - id / title: *(feel free to add more courses yourself!*
   DS1337 / Cool design
   PG1011 / Binary programming
   PJ1234 / Basic project

8. Register 2 fields of study: Dynamic Design and Pragmatic Programming.
   Make connections between fields of study and courses so that the result is:
   - Pragmatic Programming contains the courses PJ1234 and PG1011.
   - Dynamic Design contains the courses PJ1234 and DS1337.

9. Register yourself as a student in one of the field of studies.

10. Create a query to find which courses you're following, based on your name.

Task 3 – Modelling task from UCF

As an extra Modelling task set, solve this one from the University of Central Florida:
https://www.cs.ucf.edu/courses/cop4710/fall2013/hw1KEY%20-%20new.pdf
(Solutions are provided in the pdf, just after each question.)

It includes usage of today's extra ER diagram symbols (named "participation"), and the first task is a "reverse diagram"-thingy where you see the diagram and then explain what it means. 😊

Task 4 – Even more modelling

This is the end of the modelling topic & exercises. But it will be a part of the exam!

So, either:

- Look online for more modelling tasks that you can solve on your own.
- Or think about what domain you want to work on for your exam hand-in (a favourite hobby, or the like) and start planning an ER model for that domain.